Navigation Product Analytics

Route Recommendation & User Behavior — Business Case

Prepared for: Product Analytics Candidate / Portfolio Project

Prepared by: Product Analytics Team, PathWise Technologies

Document type: Case Study Brief & Analytics Assignment

Classification: Internal — Practice Dataset (Synthetic)

1. Company & Product Background

Company

PathWise Technologies is a mid-size mobility-software company that builds a consumer navigation and route-recommendation app used for everyday driving: commuting, errands, and road trips. PathWise is not a ride-hailing or delivery company — it is a pure navigation product that competes on route quality, ETA accuracy, and trust.

Product Description

The PathWise app lets a user search for a destination and receive a small set of candidate routes (typically two to three), each generated by PathWise's routing engine along a different optimization axis — fastest, cheapest (least toll cost), balanced, toll-free, or scenic. The engine ranks one route as the primary recommendation. The user can accept that recommendation or manually select an alternative before starting turn-by-turn navigation.

Target Users

- Daily commuters in mid-size and large metro areas across North America, Europe, Australia, and India.
- Occasional trip planners (weekend errands, travel, unfamiliar destinations).
- A smaller base of professional/frequent drivers (delivery contractors, field-service workers) who generate a disproportionate share of sessions.

Business Model

PathWise operates on a freemium subscription model: a free ad-supported tier (**Free**), a mid-tier subscription with offline maps and fewer ads (**Plus**), and a top-tier subscription with advanced routing, live traffic overlays, and premium support (**Premium**). Retention and engagement directly influence subscription upgrades and ad revenue.

Product Scale

- Roughly 20,000 monthly active users in the current analytics extract (a sampled slice of the full production user base, sized for this case study).
- Hundreds of thousands of navigation sessions per period across 15 metro areas in 6 countries.
- Millions of in-app funnel events captured for behavioral analysis.

Main Product Features

- Destination search with recent/favorite destinations.
- Multi-route recommendation (Fastest, Cheapest, Balanced, Toll-Free, Scenic).
- Turn-by-turn navigation with live rerouting when conditions change.
- Traffic- and weather-aware ETA prediction.
- A/B-tested experimentation framework for route-presentation changes.

Current Business Situation

PathWise has healthy top-of-funnel growth (destination searches are up year over year), but leadership has flagged that a meaningful share of users do not accept the app's top route recommendation, and that navigation abandonment and 30-day retention have plateaued. The Product and Growth teams have jointly requested an analytics investigation ahead of the next roadmap planning cycle.

2. Business Problem

Users frequently reject the route recommended by PathWise's routing algorithm in favor of a manually selected alternative. This matters because the recommendation is the product's core value proposition: if users routinely distrust or override it, PathWise is effectively acting as a plain map, not an intelligent navigation assistant.

Why route-recommendation acceptance matters

- Acceptance is a direct signal of algorithmic trust — low acceptance suggests the ranking model is not capturing what users actually value.
- Every manual override adds friction to the search-to-navigation funnel and increases the chance the user abandons the session entirely.
- Acceptance is the input the recommendation model would need in order to personalize and improve over time.

Why users may reject recommended routes (open questions, not conclusions)

- Do users distrust the predicted ETA, especially under uncertain traffic or weather conditions?
- Do users have route-type preferences (e.g., avoiding tolls, preferring scenic routes) that the current ranking does not weight enough?
- Does route complexity (number of turns, unfamiliar roads) discourage acceptance regardless of ETA?
- Do new vs. returning users behave differently as they build trust in the product over time?

Why this problem is important for the business

- Route rejection may be contributing to navigation abandonment and lower completion rates, both of which are visible, trackable product-quality metrics.
- Poor navigation experiences plausibly affect short- and long-term retention, which ties directly to subscription conversion and ad revenue.
- Understanding the drivers of rejection lets Product prioritize the highest-leverage fix — better ETA communication, better ranking, or better personalization — rather than guessing.

Decisions the Product team needs to make

- Should the routing/ranking model be re-weighted, and on which signals?
- Should the app change how it communicates ETA uncertainty to the user (e.g., ranges instead of a single number)?
- Should route recommendations be personalized per user or per segment?
- Where in the funnel should engineering and design effort be prioritized first?

This document intentionally does not answer these questions. The analyst is expected to investigate the accompanying dataset and reach evidence-based conclusions.

3. Product Funnel

The core navigation funnel that this analysis is built around:

- **Destination Search** — the user opens the app and searches for or selects a destination.
- **Route Recommendations Shown** — the routing engine returns a ranked set of route options (typically 3) and the user views them.
- **Route Selected** — the user taps a route to proceed with, either the top recommendation (an *acceptance*) or an alternative (a *rejection*).
- **Navigation Started** — the user taps 'Start' and turn-by-turn guidance begins.
- **Navigation Completed** — the trip runs to completion without the user abandoning the session.
- **Destination Reached** — the app confirms arrival at the destination.

Each stage can be measured independently (stage-to-stage conversion) or as a cumulative funnel from search to arrival. Drop-off can occur at any stage, and different drivers likely dominate different stages — for example, ETA distrust may drive drop-off at Route Selected, while poor live conditions may drive drop-off between Navigation Started and Navigation Completed.

4. Business Objectives

1. **Improve recommended-route acceptance** — increase the share of sessions where users accept the algorithm's top recommendation.
2. **Improve ETA accuracy** — reduce the gap between predicted ETA and actual trip duration.
3. **Reduce navigation abandonment** — lower the share of started navigations that end in session_abandoned rather than completion.
4. **Reduce unnecessary rerouting** — minimize mid-trip reroutes that are not justified by genuine route disruptions.
5. **Improve navigation completion** — increase the share of started trips that finish successfully.
6. **Improve 30-day user retention** — increase the share of users who remain active a month after signup or after a given session.
7. **Personalize route recommendations** — tailor the ranked route list to each user's demonstrated preferences (route type, toll tolerance, etc.).

5. Key Product Questions

The analyst is expected to answer the following questions using the accompanying dataset. No answers are provided in this document.

Funnel

- What percentage of destination searches result in a route being selected?
- What is the overall recommended-route acceptance rate, and how does it vary by segment?
- Where is the largest drop-off in the funnel, and does it differ by city, device, or user type?
- What share of started navigations are abandoned before completion?

User Behavior

- Do new users behave differently from returning users in terms of acceptance and completion?
- Which user segments (city, subscription tier, device, preferred route type) have the highest navigation frequency?
- Which users are most likely to reject recommended routes, and what do they have in common?
- How does session activity vary by time of day and day of week?

Route Behavior

- Does toll presence on the recommended route influence acceptance?
- Does traffic level or traffic-uncertainty score influence route rejection?
- Does route complexity (turns, complexity score) influence user choice?
- Why might users sometimes select a slower or longer route over the recommended one?
- How does road quality relate to acceptance and completion?

ETA

- How accurate are recommended ETAs, overall and by traffic/weather condition?
- Does ETA error increase the likelihood of route rejection?
- Which cities or time periods show the worst ETA accuracy?
- Is ETA error systematically biased (consistently under- or over-predicting), or symmetric noise?

Retention

- Does a poor navigation experience (abandonment, heavy rerouting) affect 7-day or 30-day retention?
- Do users with frequent rerouting return to the app less often than users with smooth trips?
- How do DAU, WAU, and MAU trend over the observation window?

Product Strategy

- What product changes should be prioritized based on the evidence?
- Should route ranking be personalized, and for which segments would it matter most?
- Should the app display an ETA confidence range instead of a single-number ETA?
- What is the estimated business impact of the recommended changes?

6. KPI Framework

Acquisition

KPI	Definition	Formula	Business Meaning
New Users	Users who signed up in a given period	COUNT(DISTINCT user_id WHERE signup_date in period)	Top-of-funnel growth
Destination Searches	Total destination-search events in a period	COUNT(event_type = 'destination_search')	Demand / intent volume

Activation

KPI	Definition	Formula	Business Meaning
First Navigation	Share of new users who start at least one navigation	COUNT(DISTINCT users w/ navigation_started) / COUNT(new users)	Whether new users reach real usage
Search-to-Navigation Conversion	Share of destination searches that lead to a started navigation	COUNT(navigation_started) / COUNT(destination_searched)	Destination-search efficiency

Engagement

KPI	Definition	Formula	Business Meaning
Navigation Sessions / User	Average sessions per active user in a period	COUNT(sessions) / COUNT(DISTINCT active users)	Depth of engagement
Weekly Active Navigators (WAN)	Distinct users with >=1 navigation session in a 7-day window	COUNT(DISTINCT user_id, 7-day window)	Short-term engagement health
Monthly Active Navigators (MAN)	Distinct users with >=1 navigation session in a 30-day window	COUNT(DISTINCT user_id, 30-day window)	Medium-term engagement health

Product Quality

KPI	Definition	Formula	Business Meaning
Route Acceptance Rate	Share of sessions where the selected route is the recommended route	COUNT(selected_route_id = recommended_route_id) / COUNT(sessions)	Core product quality
ETA Accuracy	How close predicted ETA is to actual duration	1 - AVG(ABS(actual_duration_minutes - recommended_eta_minutes) / actual_duration_m	Model prediction quality
ETA Error	Average absolute or signed prediction error	AVG(actual_duration_minutes - recommended_eta_minutes)	Directional over-/under-prediction bias
Reroute Rate	Average reroutes per started navigation	AVG(reroute_count WHERE navigation_started = true)	Navigation stability
Navigation Completion Rate	Share of started navigations that complete successfully	COUNT(navigation_completed) / COUNT(navigation_started)	End-to-end reliability of the navigation experie

Retention

KPI	Definition	Formula	Business Meaning
7-Day Retention	Share of a cohort active again within 7 days	COUNT(active in day 1–7) / COUNT(cohort)	Short-term stickiness
30-Day Retention	Share of a cohort active again within 30 days	COUNT(active in day 1–30) / COUNT(cohort)	Longer-term product-market fit signal

7. Hypotheses

The analyst should treat each of the following as testable hypotheses, not established facts.

H1: Higher ETA error increases the probability of route rejection.

- **Null (H0):** ETA error has no relationship with rejection probability.
- **Alternative (H1):** ETA error is positively associated with rejection probability.
- **Suggested test:** Logistic regression / point-biserial correlation

H2: Users are more likely to reject routes that include tolls.

- **Null (H0):** Toll presence has no effect on acceptance rate.
- **Alternative (H1):** Toll presence reduces acceptance rate.
- **Suggested test:** Two-proportion z-test / chi-square test of independence

H3: High traffic increases the likelihood of alternative-route selection.

- **Null (H0):** Traffic level has no effect on route selection behavior.
- **Alternative (H1):** Higher traffic level increases alternative-route selection.
- **Suggested test:** Chi-square test of independence

H4: Frequent rerouting reduces navigation completion rate.

- **Null (H0):** Reroute count has no relationship with completion.
- **Alternative (H1):** Higher reroute count is associated with lower completion probability.
- **Suggested test:** Logistic regression / Mann-Whitney U test

H5: Poor navigation experiences (abandonment) negatively affect subsequent user retention.

- **Null (H0):** Navigation outcome has no effect on 7-/30-day retention.
- **Alternative (H1):** Users with abandoned sessions show lower subsequent retention.
- **Suggested test:** Two-proportion z-test / survival analysis

H6: Returning users are more likely to accept recommended routes than new users.

- **Null (H0):** Acceptance rate is equal between new and returning users.
- **Alternative (H1):** Returning users show a different (likely higher) acceptance rate.
- **Suggested test:** Two-proportion z-test

H7: Route complexity (turns, complexity score) reduces acceptance likelihood.

- **Null (H0):** Route complexity has no relationship with acceptance.

- **Alternative (H1):** Higher complexity is associated with lower acceptance.
- **Suggested test:** Logistic regression / correlation analysis

H8: ETA accuracy varies significantly by city.

- **Null (H0):** ETA error is equal across cities.
- **Alternative (H1):** ETA error differs across at least one pair of cities.
- **Suggested test:** One-way ANOVA / Kruskal-Wallis test

H9: Displaying an ETA confidence range increases route-recommendation acceptance (A/B test).

- **Null (H0):** Acceptance rate is equal between Control and Treatment groups.
- **Alternative (H1):** Acceptance rate differs between Control and Treatment groups.
- **Suggested test:** Two-proportion z-test (A/B test)

8. A/B Testing Scenario

Product hypothesizes that showing users a range-based ETA with an explicit uncertainty message, instead of a single confident number, will increase trust and therefore route acceptance.

Control

Standard single-number ETA: **“Fastest route — 32 min”**

Treatment

Range-based ETA with uncertainty messaging: **“32–40 min — High traffic uncertainty”**

The analyst must determine, using the *experiments.csv* table, whether showing an ETA confidence range improves recommended-route acceptance, and what (if any) trade-offs appear on secondary and guardrail metrics.

Primary Metric

Recommended-route acceptance rate (route_acceptance)

Secondary Metrics

- Navigation completion rate
- Reroute count / rate
- Session abandonment rate
- 7-day retention

Experiment Design Parameters

Parameter	Value / Guidance
Experiment duration	4–6 weeks, sufficient to reach the minimum detectable effect at the observed baseline conversion rate
Control / Treatment structure	50% Control / 50% Treatment, randomized at exposure
Randomization unit	User (user_id) — each user is consistently assigned to one arm for the duration of the test to avoid within-user
Sample-size considerations	Powered against the observed baseline acceptance rate and a minimum detectable effect the Product team co
Success criteria	Statistically significant lift in primary metric with no statistically significant regression on guardrail metrics
Guardrail metrics	Session abandonment rate; navigation completion rate should not regress

Experiment results and conclusions are intentionally not provided — the analyst should derive them from the experiments.csv dataset.

9. Expected Deliverables

As part of this portfolio project, produce the following:

1. SQL analysis answering the funnel, route, ETA, and retention questions above.
2. Python analysis (EDA, statistical testing, and modeling) supporting the same questions.
3. A documented product funnel with stage-by-stage conversion rates.
4. A user segmentation analysis (behavioral and/or demographic).
5. An ETA accuracy analysis, including error distribution and any systematic bias.
6. A retention / cohort analysis (7-day and 30-day, by signup cohort and by experience quality).
7. An A/B test analysis of the ETA-confidence-range experiment.
8. A 4-page Power BI dashboard covering the executive overview, route analytics, user behavior/retention, and experiment insights.
9. A set of product recommendations grounded in the analysis.
10. An executive summary suitable for a non-technical product leadership audience.